

portance of further studies on the neutrino mass should not be underestimated. The most promising theoretical option is that the mass type is exactly the one proposed by Majorana. Its experimental demonstration is a concrete and well-defined goal to strive for in the exploration of physics beyond the Standard Model.

The best way to probe the Majorana nature of neutrinos is to measure the rate of neutrinoless double beta decay — i.e., the rate at which electron pairs are created in certain nuclear decays — an observation which would lead to a profound change in our understanding of matter. Although we do not have an established theory that can guide us safely in the next steps, we are just starting a pioneering exploration of the next two, uncharted orders of magnitude, an exciting journey that will bring us a step closer to unlock the secrets of the universe.

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